Ansible Configuration Management Project Submission

Project Summary

Project: Multi-Environment Infrastructure Automation with Ansible
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GitHub Repository: StegHub_DevOps/project12 at main · 123Origami/StegHub_DevOps
LinkedIn Post: https://www.linkedin.com/posts/sally-m-09546a245_ansible-devops-configurationmanagement-activity-7398438705025376256-gpLz?utm_source=share&utm_medium=member_desktop&rcm=ACoAADzCKfEBZK3LPTQ5D0pi879scjlAX0g4FTk

Project Overview

Built a complete Ansible configuration management system supporting dynamic assignments, community role integration, and conditional multi-environment deployments.

Core Components Delivered

- **Dynamic Configuration Management** - Environment-specific variable loading
- **Community Role Integration** - MySQL, Apache, PHP from Ansible Galaxy
- **Conditional Load Balancers** - Nginx/Apache deployment based on environment variables
- **Multi-Environment Infrastructure** - Dev, UAT, Production configurations
- **Git Workflow Implementation** - Feature branching and PR process

Key Features Implemented

- Smart environment detection and variable loading
- Load balancer selection (Nginx vs Apache) via configuration
- Database automation with MySQL setup and user management
- Multi-server coordinated deployment
- Production-ready project structure

Skills Demonstrated

Technical Competencies

- Ansible playbook development and organization
- YAML configuration management

- Community role integration and customization
- Multi-environment deployment strategies
- Infrastructure as Code principles
- Git version control workflows

Problem-Solving Achievements

- Resolved complex YAML syntax and dependency issues
- Troubleshot SSH connectivity and authentication challenges
- Implemented conditional deployment logic
- Managed environment-specific configurations
- Established scalable project structure

Validation & Testing

Quality Assurance

- Syntax validation of all playbooks and roles
- Dry-run execution testing across environments
- Environment-specific deployment verification
- Load balancer switching functionality testing
- Database creation and permission validation
- Multi-server connectivity confirmation

Success Criteria Met

- Dynamic variable assignment operational across environments
- Community roles properly integrated and configured
- Conditional load balancer deployment functional
- Multi-environment support fully operational
- Git workflow with proper branching and merging
- Comprehensive documentation completed

Future Enhancement Opportunities

- Ansible Vault implementation for sensitive data
- CI/CD pipeline integration
- Comprehensive testing suite development
- Monitoring and logging configuration
- Automated backup and rollback procedures